



Takoua Khammassi

Software Engineering Student — Data Science & AI

☎ +216 93 805 222 ✉ takoua.khammassi@isamm.u-manouba.tn in [takoua-khammassi](#) 🌐 [takouakhammassi](#)
🌐 <https://my-portfolio-takoua.vercel.app/> 📍 Tunis, Tunisia

PROFILE

Final-year software engineering student at ISAMM, specializing in Data Science and AI through internships and personal projects in Machine Learning, Deep Learning, and Computer Vision. I am currently seeking a final-year internship (PFE) in France to contribute to innovative projects in this field, while further developing my technical skills and thriving in a stimulating international environment.

TECHNICAL SKILLS

Programming Languages : Python, SQL, JavaScript

Data Science & AI : Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, Machine Learning, Deep Learning, Transfer Learning, Transformers, Computer Vision, NLP, LLMs, RAG

Big Data : Hadoop, HDFS, Spark, NoSQL

Tools : Git, Docker, Streamlit

Visualization : Power BI, Matplotlib, Seaborn

EXPERIENCE

AI & Machine Learning Intern — Bee Coders

July 2026 – August 2026

Stack : Python, Flask, Symphony, MySQL, NLP

- Cleaned 5 heterogeneous data sources and benchmarked 3 NLP approaches (TF-IDF, fuzzy matching, multilingual embeddings) for intent recognition
- Built a Flask API and a Symphony web application with secure authentication and multi-conversation chat

Software Engineering Intern — Sagemcom

August 2025 – Sept. 2025

Stack : Python, Pandas, NumPy, Matplotlib

- Built a Python application automating Wi-Fi test analysis in an industrial environment
- Designed statistical visualizations and a process capability analysis to validate RF measurement compliance

AI & Data Science Intern — DeepAI

Feb. 2024 – May 2024

Stack : Python, SQL Server, ETL, Power BI

- Built an AI pipeline (web scraping/crawling, OCR, HTML/PDF/video cleaning, Zero-Shot classification, K-means clustering) processing 193,000+ words of educational resources
- Designed a SQL Server data warehouse and Power BI dashboards analyzing 874,000 characters of content

PROJECTS

Hirfatuna — AI Platform for Tunisian Craft Valorization

2026

Personal project — Stack : Python, PyTorch, ResNet-50, Qwen3.5-VL, FAISS, RAG, Streamlit

- Built an AI platform combining computer vision and generative AI to identify Tunisian handicrafts
- Built a 3,264-image dataset and trained a ResNet-50 model reaching 90.8% accuracy
- Integrated a RAG chatbot (Qwen3.5-VL, FAISS) and deployed the application to production with Streamlit

HDFS Small File Optimization

2026

Research project (PFA) — Stack : Python, Hadoop, HDFS

- Developed an HDFS small-file merging strategy based on a modified Louvain algorithm
- Improved storage efficiency by reducing NameNode memory usage by 64.7% and HDFS blocks by 97.1%

EDUCATION

Engineering Cycle in Computer Science and Multimedia — Software Engineering

[2024 – Present]

Higher Institute of Multimedia Arts of Manouba (ISAMM)

Bachelor's Degree in Big Data and Data Analysis

[2021 – 2024]

Higher Institute of Multimedia Arts of Manouba (ISAMM)

CERTIFICATIONS

- Associate Data Scientist — DataCamp (2026)
- Data Science Essentials with Python & Introduction to Data Science — Cisco Networking Academy (2026)
- Data Science & Analytics — HP Foundation (2026)
- AI Fundamentals — Cisco Networking Academy & IBM (2024)
- Deep Learning using TensorFlow — IBM (2024)